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Zipf Distribution

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Zipf distributions are used to sample data based on zipf's law.

Zipf's Law: In a collection, the nth common term is $1/n$ times of the most common term. E.g. the 5th most common word in English occurs nearly $1/5$ times as often as the most common word.

It has two parameters:

a - distribution parameter.

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NumPy Random

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NumPy ufunc

Example

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Draw out a sample for zipf distribution with distribution parameter 2 with size 2x3:

```
from numpy import random

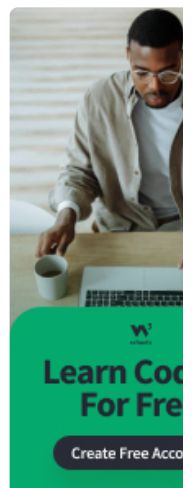
x = random.zipf(a=2, size=(2, 3))

print(x)
```

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Visualization of Zipf Distribution

Sample 1000 points but plotting only ones with value < 10 for more

[COLOR PICKER](#)

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Quiz/Exercises

meaningful chart.

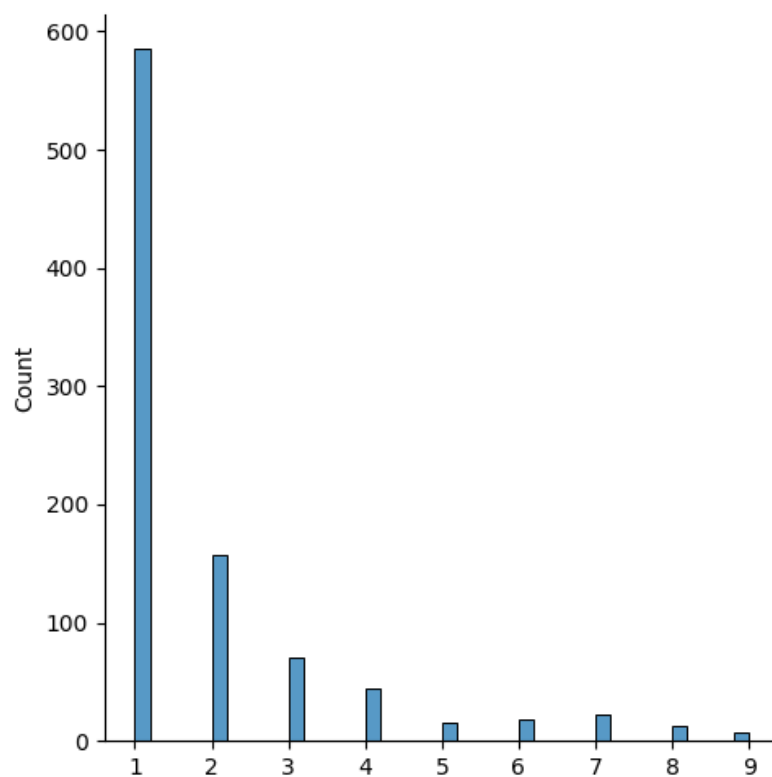
Example

```
from numpy import random
import matplotlib.pyplot as plt
import seaborn as sns

x = random.zipf(a=2, size=1000)
sns.displot(x[x<10])

plt.show()
```

Result



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Exercise ?

How many parameters does the `random.zipf()` method have?

○ 1



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